TrafficPolice: Machine Learning Based Network Intrusion Detection

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***Abstract*—TrafficPolice is a machine learning based network intrusion detection system developed using the UNSW-NB15 dataset. The project compares four base models: Logistic Regression, Random Forest, Isolation Forest, and One-Class SVM. Random Forest with SMOTE achieved the best base F1-score and was further tuned using Optuna. The final tuned model achieved 90.06% accuracy, 99.00% precision, 86.28% recall, and 92.20% F1-score. The system was deployed using FastAPI, Docker, Amazon ECR, and AWS Elastic Beanstalk with a CSV-upload dashboard.**

***Index Terms*—Intrusion Detection, UNSW-NB15, Random Forest, SMOTE, Optuna, FastAPI, Docker, AWS**

I. Introduction

Network intrusion detection systems are used to identify malicious network traffic and protect systems from attacks. Traditional rule-based systems may fail against new or changing attack patterns. Therefore, this project uses machine learning to classify network traffic as *Normal* or *Attack*. The aim was to build a complete pipeline including preprocessing, model comparison, hyperparameter optimization, evaluation, and cloud deployment.

II. Dataset and Preprocessing

The UNSW-NB15 dataset was used because it contains realistic normal and attack network traffic. It includes 82,332 training records and 175,341 testing records. After removing `id` and `attack_cat`, 42 features were used. Important categorical columns such as `proto`, `service`, and `state` were converted using target encoding. SMOTE was applied only on the training data to reduce imbalance and avoid test-data leakage.

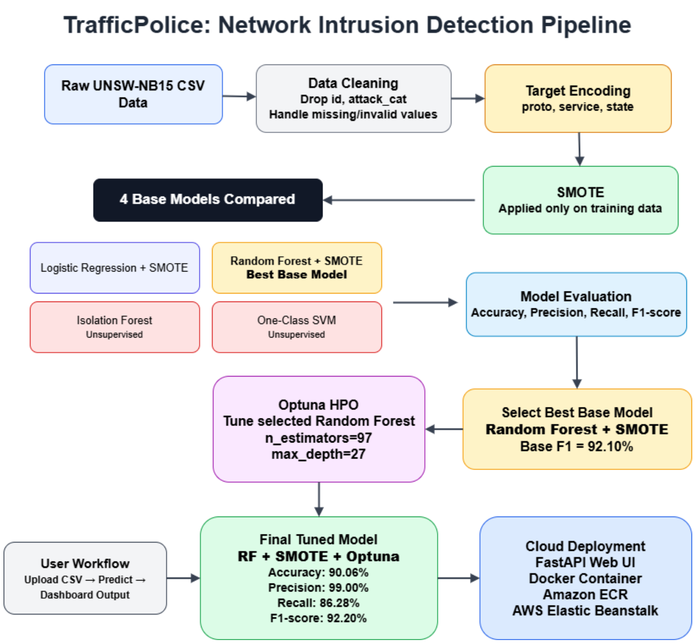

Fig. 1. TrafficPolice preprocessing, training, tuning, and deployment pipeline.

III. Methodology

Four base models were trained and compared. The supervised models were Logistic Regression + SMOTE and Random Forest + SMOTE. The unsupervised models were Isolation Forest and One-Class SVM. Model selection was based mainly on F1-score because intrusion detection requires balance between precision and recall. Random Forest + SMOTE achieved the best base F1-score, so Optuna was applied only to this selected model for hyperparameter tuning. The best tuned parameters were `n_estimators=97`, `max_depth=27`, `min_samples_split=9`, and `min_samples_leaf=1`.

IV. Results

Table I shows that Random Forest + SMOTE was the strongest base model. After Optuna tuning, the final tuned model slightly improved the F1-score from 92.10% to 92.20%.

TABLE I
Model Performance on UNSW-NB15 Test Set

Model	Acc.	Prec.	Rec.	F1
Logistic Regression + SMOTE	85.69%	97.89%	80.71%	88.47%
Random Forest + SMOTE	89.95%	99.01%	86.10%	92.10%
Isolation Forest	64.90%	80.78%	63.56%	71.14%
One-Class SVM	65.87%	81.82%	64.09%	71.88%
RF + SMOTE + Optuna	**90.06%**	**99.00%**	**86.28%**	**92.20%**

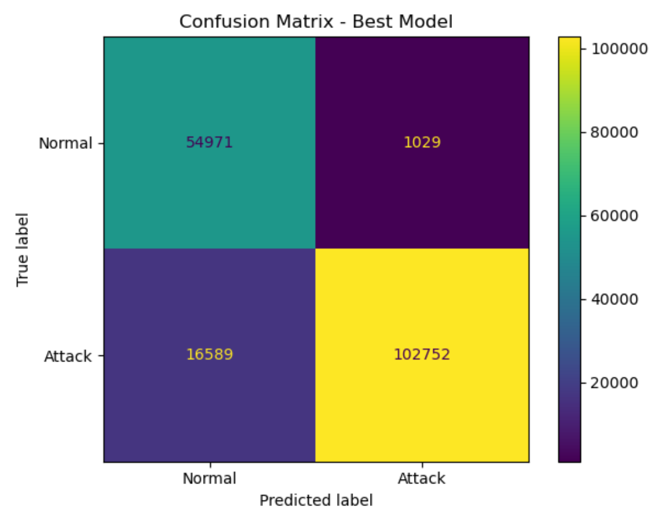

Fig. 2. Confusion matrix of the final tuned Random Forest model.

The confusion matrix shows 54,971 true normal records, 1,029 false positives, 16,589 false negatives, and 102,752 true attack. The model has high precision, predicted attacks are usually correct. Recall is lower because some attacks are still missed.

V. Deployment

The final model was integrated into a FastAPI backend with an HTML dashboard. Users can upload a CSV file and view predicted normal or attack records. The application was containerized using Docker, pushed to Amazon ECR, and deployed on AWS Elastic Beanstalk. This satisfies the requirement of a working cloud-deployed UI.

VI. Conclusion

TrafficPolice successfully demonstrates a complete machine learning pipeline for network intrusion detection. Four base models were compared first, Random Forest + SMOTE was selected as the best base model, and Optuna was then used to tune it further. The final model achieved 92.20% F1-score and was deployed as a Dockerized web application on AWS.

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